NEW publications

Code: Tj = Textbook, junior high	L = Library
Ts = Textbook, senior high	P = Professional
Tt = Textbook, two-year college	S = Supplementary student reading

Algebra—Programmed: Part 3, 2d ed. (Tt), Robert H. Alwin, Robert D. Hackworth, and Joseph W. Howland. 1980, xii + 596 pp., $15.95 paper. ISBN 0-13-021931-2. Prentice-Hall, Englewood Cliffs, NJ 07632.

This latest edition of a second half of an elementary algebra course is composed of relatively clear telegraphic explanations followed by a handful of exercises and an abundance of well-structured, self-evaluative tests.

The text covers traditional topics with a somewhat increased emphasis on exponents, polynomials, and the arithmetic of square roots. The units on sets, functions, slope, and scientific notation are particularly effective. Considerable space is devoted to logarithms, but the topic is developed in such a way that the logic is compatible with that of the hand-held calculator. Unfortunately, the authors do not envision the student using a calculator for lengthy computations, and they include a lengthy unit on arithmetic operations with logarithms.

An additional feature of this second edition is a concluding chapter devoted to applications. The problems are uninspiring and painfully familiar: boats going downstream, grocers mixing nuts, cars overtaking one another, and so on. Were the applications to be interspersed wherever appropriate and designed to keep the student interested and aware of the usefulness of the concepts studied, this aspect of the programmed text would be more effective and valuable.

Programmed texts are ultimately collections of recipes. This is one of the more readable and competent, although conventional, books.—*Myra R. Lipman, Pace University, New York, NY 10038.*

Algebra and Trigonometry (Ts, Tt), Walter Fleming and Dale Varberg. 1980, xvi + 521 + A34 + I3 pp., $17.95 cloth. ISBN 0-13-021824-3. Prentice-Hall, Englewood Cliffs, NJ 07632.

The book is instructive but not rigorous and technical. Clear and precise guidance is given for the many problems and word problems.

There is a substantial amount of reading required, and the book is not dull. The chapters are introduced with anecdotes, quotations, and historical facts. Often, the introductions are humorous. Each chapter has a summary and a review problem set. Formulas for geometry, algebra, and trigonometry are easily found on the inside covers. Answers for odd-numbered problems are given.

This is a very good book and highly recommended for use.—*Julia J. Grice, Roosevelt High School, Washington, DC 20011.*

Algebra and Trigonometry: Functions and Applications (Ts, Tt), Paul A. Foerster. 1980, 602 pp., $15.00 cloth. ISBN 0-201-02475-6. Addison-Wesley, Reading, MA 01867.

This is a unique text, rigorous in theory and rich in applications. End-of-chapter problems create mathematical models of real-world situations and employ several mathematical concepts in a single context. There are over 250 such applications in the book with titles like "Diesel Car vs. Gasoline Car Problem" (linear systems), "Louisiana Purchase Problem" (geometric series), and "Why Mammals Are the Way They Are" (joint variations); they are imparted with a sense of humor, pragmatism, and curiosity.

The book is designed for a course in intermediate algebra, advanced algebra, and trigonometry. (The three chapters on trigonometry are adapted from the author's equally impressive *Trigonometry: Functions and Applications.*) Two appendices cover a review of computer programming and mathematical induction. Many of the application problems are appropriate for calculator or computer solutions; answers to all odd-numbered problems are given.

This is a book to be read. Concepts tests completing each chapter require the student to think, write, and synthesize unfamiliar material. As a basic text, it is appropriate for the above average algebra student. As a source of application problems for all, it is peerless.—*William Jacob Bechem, Greenwich High School, Greenwich, CT 06830.*

Applied Algebra II (Tt), Thomas J. McHale and Paul T. Witzke. 1980, 444 pp., $12.95 paper. ISBN 0-201-04775-6. Addison-Wesley Publishing Co., Reading, MA 01867.

The book is in programmed learning form and was reproduced from camera-ready copy supplied by the authors. Was it field-tested? There is no mention of it in the preface or the press release. Good field-testing, with appropriate revisions, would have eliminated some awkward layout, such as the minus sign that runs into a radical and makes it look like the principal root. The book would also have been better if the authors had stuck to standard mathematical terminology, instead of introducing such concepts as *complete* and *incomplete* complex numbers. Some statements